

EPSON OPOS ADK MANUAL

**APPLICATION DEVELOPMENT
GUIDE**

Compatible Mode

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Section 1. Introduction

The specifications for the initialization, devices with hydra connection and Error Event are different in pre-Ver2.60 and Ver2.60 in compliance with UPOS1.11.

The compatibility mode for upward compatibility was added in OPOS Ver2.60.

You can operate with Ver2.50 specifications by setting the compatible mode in the "Version compatibility" dialog box of SetupPOS Utility.

For the settings of the compatibility mode, please refer to "EPSON OPOS ADK MANUAL User's Manual (Installer/ SetupPOS/ TMUSB)".

Section 2. Outline

The targets of switching specifications in this compatible mode are as follows.

- Initialization processing
- Switch-over of Hydra error code
- Error Event processing

These can operate with Ver2.50 specifications by setting the compatible mode in the "Version compatibility" dialog box of SetupPOS Utility.

2.1 Initialization processing

The timing of initialization processing is different in pre-Ver2.60 and Ver2.60.

The target categories of switching specifications are POSPrinter, LineDisplay and CashDrawer.

For details, please refer to the "Section 3 The specification of each device" of this manual.

The ElectronicJournal is operable in pre-Ver2.60 as in Ver2.60. For details, please refer to the "3.5 ElectronicJournal" of this manual.

2.2 Switch-over of Hydra error code

Error codes for devices with hydra connection are different in pre-Ver2.60 and Ver2.60.

This switchover feature is to enable switching between different methods of processing error codes in Hydra devices. The target categories of switching specifications are LineDisplay and CashDrawer.

For details, please refer to the "Section 3 The specification of each device" of this manual.

2.3 Error Event processing

The issue methods of the error event are different in pre-Ver2.60 and Ver2.60.

The target category of switching specifications is MSR.

For details, please refer to the "Section 3 The specification of each device" of this manual.

Section 3. The specification of each device

Explains about the switching specifications peculiar to each device class, in this section

3.1 POSPrinter

The differences in POSPrinter specifications of pre-Ver2.60 and Ver2.60 are as follows.

3.1.1 Initialization processing

<Method>

Method Name	Ver2.60 specifications	Ver2.50 specifications
Open	Initialize property values based on "Device Specific Settings" in the SetupPOS utility.	(The same as the Ver2.60 specifications)
Claim	—	Attempts to connect to the device, updates properties and initializes the device.
	This method is finished normally even when a device status is offline. ^{*1}	This method returns error when a device status is offline.

<Property>

Property Name	Ver2.60 specifications	Ver2.50 specifications
DeviceEnabled	After Claim method is called, initialization process is started when the first time of setting this property at TRUE.	—
	When initialization process was completed normally or when an error occurred, property is established in TRUE and returns. When an error occurs in the initialization process, process is resumed automatically when an error factor is removed. ^{*1}	—

Property Name	Ver2.60 specifications	Ver2.50 specifications
CapRecMarkFeed	When Open method is called, initializes property values based on "Device Specific Settings" in the SetupPOS utility.	(The same as the Ver2.60 specifications)
CapRecPapercut		
CapSlpBothSidesPrint		
CharacterSet		
CharacterSetList	After Claim method is called, updates properties based on the information attained from the device when the first time of setting DeviceEnabled property at TRUE.	When Claim method is called, updates properties based on the information attained from the device.
RecLineChars		
RecLineCharsList		
RecSidewaysMaxChars		
SlpLineChars		
SlpLineCharsList		
SlpSidewaysMaxChars		

*1 For serial interface, it is finished normally even when a device is Power OFF or Unconnected.

3.2 LineDisplay

The differences in LineDisplay specifications of pre-Ver2.60 and Ver2.60 are as follows.

3.2.1 Initialization processing

<Method>

Method Name	Ver2.60 specifications	Ver2.50 specifications
Open	—	—
Claim	Not attempts to connect to the device and not set property.	Attempts to connect to the device and set property.
	Not redraw the marquee.	Redraw the marquee.
	This method is finished normally even when a device status is offline. ^{*1}	This method returns error when a device status is offline.

<Property>

Property Name	Ver2.60 specifications	Ver2.50 specifications
DeviceEnabled	Initialization process is started asynchronously when this property established in TRUE.	—
	Marquee process is not started when this property established in TRUE.	Marquee process is started when this property established in TRUE.
	When initialization process was completed normally or when an error occurred, property is established in TRUE and returns. When an error occurs in the initialization process, process is resumed automatically when an error factor is removed. When initialization process was completed normally, marquee process is started.	—

^{*1} For serial interface, it is finished normally even when a device is Power OFF or Unconnected.

3.2.2 Switch-over of Hydra error code

Content of error	Ver2.60 specifications	(ResultCode)
		(ResultCodeExtended)
Content of error	Ver2.60 specifications	Ver2.50 specifications
Cover Open	E_FAILURE EPTR_COVER_OPEN	E_OFFLINE EPTR_COVER_OPEN
No Journal paper	E_FAILURE EPTR_JRN_EMPTY	E_OFFLINE EPTR_JRN_EMPTY
No Receipt paper	E_FAILURE EPTR_REC_EMPTY	E_OFFLINE EPTR_REC_EMPTY
Cartridge removed	E_FAILURE EPTR_REC_CARTRIDGE_REMOVED	E_OFFLINE EPTR_REC_CARTRIDGE_REMOVED
Replace Ink cartridge	E_FAILURE EPTR_REC_CARTRIDGE_EMPTY	E_OFFLINE EPTR_REC_CARTRIDGE_EMPTY
Head cleaning of Printer in execution	E_FAILURE EPTR_REC_HEAD_CLEANING	E_OFFLINE EPTR_REC_HEAD_CLEANING
A label paper jam occurred within the peeler system	E_FAILURE EPTR_LABEL_JAM	E_FAILURE EPTR_LABEL_JAM
Mechanical error	E_FAILURE EPTR_MECHANICAL	E_FAILURE EPTR_MECHANICAL
Cutter error	E_FAILURE EPTR_CUTTER	E_FAILURE EPTR_CUTTER
Irrecoverable error	E_FAILURE EPTR_UNRECOVERABLE	E_FAILURE EPTR_UNRECOVERABLE
Automatic recoverable error	E_FAILURE EPTR_AUTORECOVERABLE	E_FAILURE EPTR_AUTORECOVERABLE

3.3 CashDrawer

The differences in CashDrawer specifications of pre-Ver2.60 and Ver2.60 are as follows.

3.3.1 Initialization processing

<Method>

Method Name	Ver2.60 specifications	Ver2.50 specifications
Open	Initialize property values based on "Device Specific Settings" in the SetupPOS utility.	(The same as the Ver2.60 specifications)
Claim	Not attempts to connect to the device and not set property.	Attempts to connect to the device and set property.
	This method is finished normally even when a device status is offline. ^{*1}	This method returns error when a device status is offline.

<Property>

Property Name	Ver2.60 specifications	Ver2.50 specifications
DeviceEnabled	After Claim method is called, initialization process is started when the first time of setting this property at TRUE.	—
	Initialization process is started when this property established in TRUE. Attempts to connect to the device asynchronously, updates properties based on the information attained from the device and initializes the device. When an error occurs in the initialization process, process is resumed automatically when an error factor is removed.	When value is established in TRUE from FALSE, attempts to connect to the device, updates properties based on the information attained from the device.
DarwerOpened	Attempts to connect to the device during initialization process, attains Drawer status and updates properties.	Attempts to connect to the device during executing Claim method, attains Drawer status and updates properties.

^{*1} For serial interface, it is finished normally even when a device is Power OFF or Unconnected.

3.3.2 Switch-over of Hydra error code

Content of error	Ver2.60 specifications	(ResultCode) (ResultCodeExtended)
		Ver2.50 specifications
Cover Open	E_FAILURE EPTR_COVER_OPEN	E_OFFLINE EPTR_COVER_OPEN
No Journal paper	E_FAILURE EPTR_JRN_EMPTY	E_OFFLINE EPTR_JRN_EMPTY
No Receipt paper	E_FAILURE EPTR_REC_EMPTY	E_OFFLINE EPTR_REC_EMPTY
Cartridge removed	E_FAILURE EPTR_REC_CARTRIDGE_REMOVED	E_OFFLINE EPTR_REC_CARTRIDGE_REMOVED
Replace Ink cartridge	E_FAILURE EPTR_REC_CARTRIDGE_EMPTY	E_OFFLINE EPTR_REC_CARTRIDGE_EMPTY
Head cleaning of Printer in execution	E_FAILURE EPTR_REC_HEAD_CLEANING	E_OFFLINE EPTR_REC_HEAD_CLEANING
A label paper jam occurred within the peeler system	E_FAILURE EPTR_LABEL_JAM	E_FAILURE EPTR_LABEL_JAM
Mechanical error	E_FAILURE EPTR_MECHANICAL	E_FAILURE EPTR_MECHANICAL
Cutter error	E_FAILURE EPTR_CUTTER	E_FAILURE EPTR_CUTTER
Irrecoverable error	E_FAILURE EPTR_UNRECOVERABLE	E_FAILURE EPTR_UNRECOVERABLE
Automatic recoverable error	E_FAILURE EPTR_AUTORECOVERABLE	E_FAILURE EPTR_AUTORECOVERABLE

3.4 MSR

The differences in MSR specifications of pre-Ver2.60 and Ver2.60 are as follows.

3.4.1 Error Event

DataEventEnabled=false、DataCount=0、Reading of the correct data card

<Property>

Property Name	Ver2.60 specifications	Ver2.50 specifications
DataCount	Updates to 1 when the data card is read.	(The same as the Ver2.60 specifications)
State	Not updates. (Still S_IDLE)	(The same as the Ver2.60 specifications)

<Event>

Event Name	Ver2.60 specifications	Ver2.50 specifications
DataEvent	Events are queued.	(The same as the Ver2.60 specifications)

DataEventEnabled=false、DataCount=0、Reading of the incorrect data card

<Property>

Property Name	Ver2.60 specifications	Ver2.50 specifications
DataCount	Not updates when the data card is read.	Updates to 1 when the data card is read.
State	When the data card is read, this property is established in S_ERROR from S_IDLE.	Not updates. (Still S_IDLE)
	When an ErrorEvent is fired, this property is established in S_IDLE from S_ERROR.	

<Event>

Event Name	Ver2.60 specifications	Ver2.50 specifications
DataEvent	Events are not queued.	(The same as the Ver2.60 specifications)
ErrorEvent	Following events are queued. ErrorCode: E_ILLEGAL (or) E_EXTENDED * ² ErrorCodeExtended: Illegal card error * ² ErrorLocus: EL_INPUT ErrorResponse: ER_CLEAR	(The same as the Ver2.60 specifications)
	When an Event is fired, updates properties based on the data read from the card. * ¹	Not updates properties when the event is fired.

*¹ The following properties of the card data are renewed.

Whether these properties are renewed is decided by the value of the ParseDecodeData property and the TracksToRead property.

TrackXData (X = 1~4)

AccountNumber

ExpirationData

Title

FirstName

MiddleInitial

Surname

Suffix

ServiceCode

TrackXDictionaryData (X = 1,2)

*² The value of the ErrorCode is changed by the value of the ErrorReportingType property.

DataEventEnabled=false、DataCount=1、Reading of the data card

There is not any difference between operations in data card.

<Property>

Property Name	Ver2.60 specifications	Ver2.50 specifications
DataCount	When the second card is read, this property is established in 0 from 1.	(The same as the Ver2.60 specifications)
State	When the second card is read, this property is established in S_ERROR from S_IDLE.	Not updates when the second card is read. (Still S_IDLE)
	When an ErrorEvent is fired, this property is established in S_IDLE from S_ERROR.	When an ErrorEvent is fired, this property is established in S_ERROR from S_IDLE.

<Event>

Event Name	Ver2.60 specifications	Ver2.50 specifications
DataEvent	Events are not queued.	(The same as the Ver2.60 specifications)
ErrorEvent	Following events are queued. ErrorCode: E_FAILURE ErrorCodeExtended: 0 ErrorLocus: EL_INPUT ErrorResponse: ER_CLEAR	(The same as the Ver2.60 specifications)
	When an Event is fired, updates properties based on the data read from the second card. ^{*1}	Not updates properties when the event is fired.

^{*1} The following properties of the card data are renewed.

Whether these properties are renewed is decided by the value of the ParseDecodeData property and the TracksToRead property.

TrackXData (X = 1~4)

AccountNumber

ExpirationData

Title

FirstName

MiddleInitial

Surname

Suffix

ServiceCode

TrackXDictionaryData (X = 1,2)

3.5 ElectronicJournal

The Electronic Journal category was supported since Version 2.60. Therefore, there is no setting to switch compatibility mode such as other device categories. Whichever of the Ver2.50 specifications and the Ver2.60 specifications is chosen, SO does the same movement.

Section 4. Warnings

- Although property initialization in Open method is a newly added feature in Ver2.60, compatibility mode allows switching back to the pre-Ver2.60 operation.
- When DeviceEnabled property is established in TRUE in offline status of device, the acquisition of status of the device may take time by the interface.
- For serial interface, Power OFF/Unconnected cannot be distinguished from OFFLINE. For the reason, the Claim method and the setting of the DeviceEnabled property are finished normally even when a device is Power OFF/Unconnected.